

Motion-System



Motion System

Arena Design System (ADS)

Version: 1.0

Status: Draft

Motion Engine:

- Framer Motion
- GSAP
- React Three Fiber
- Three.js
- Lottie
- CSS Transitions

Overview

Motion is a core part of the ArenaMind AI experience.

It communicates:

- Intelligence
- Context
- Hierarchy
- System Status
- AI Activity
- User Intent

Motion should never exist purely for decoration.

Every animation must answer at least one question:

- What changed?
 - Why did it change?
 - Where did it come from?
 - Where is it going?
-

Motion Philosophy

ArenaMind AI follows one simple principle.

| Motion should make complex systems feel understandable.

Users should never wonder:

- Where did this panel come from?
- What just happened?
- Why did this notification appear?

Every transition should provide context.

Motion Design Principles

1. Functional Motion

Every animation serves a purpose.

- ✓ Explain
 - ✓ Guide
 - ✓ Confirm
 - ✗ Distract
 - ✗ Entertain
-

2. Physical Motion

Objects behave like real objects.

Cards have weight.

Panels slide naturally.

Floating objects ease into position.

No robotic movement.

3. Continuous Motion

The application should always feel alive.

Examples

- Live heatmaps
- Flowing crowd particles
- Rotating stadium
- AI pulse
- Live dashboards
- Weather animation
- Notification streams

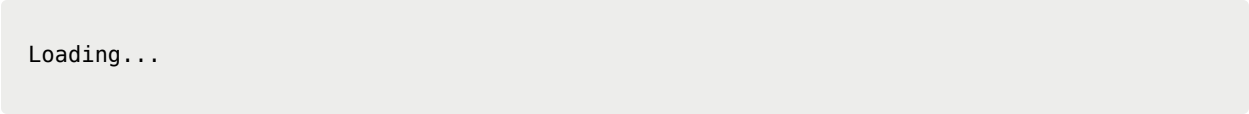
Never allow the interface to appear static.

4. AI is Alive

Artificial Intelligence should feel active.

Examples

Instead of



Loading...

Show

- Thinking pulse

- Neural network animation
 - Streaming text
 - Dynamic reasoning timeline
 - Live confidence indicator
-

Motion Hierarchy

Motion priority defines where user attention should go.

Level	Purpose
Level 1	Critical alerts
Level 2	AI responses
Level 3	Navigation
Level 4	Data updates
Level 5	Hover interactions
Level 6	Ambient motion

Motion Tokens

Duration

Token	Duration
instant	0ms
fast	150ms
normal	250ms
medium	400ms
slow	600ms
cinematic	900ms

Delay

Token	Value
none	0ms
short	100ms
medium	200ms
long	300ms

Easing

Standard

```
ease-out
```

Smooth

```
ease-in-out
```

Spring

```
type: spring  
  
stiffness: 120  
  
damping: 18
```

Never use linear animations.

Animation Categories

```
Motion  
|  
├ Page  
├ Navigation  
├ Cards  
├ AI  
└ Maps
```

- └ Digital Twin
- └ Notifications
- └ Charts
- └ Buttons
- └ Inputs
- └ Loading
- └ Background

Page Transitions

Every page transition should feel spatial.

Instead of:

Page disappears

↓

Next page appears

Use

Camera moves

↓

Page slides

↓

Content fades

↓

AI widgets appear

Duration

600ms

Navigation Motion

Sidebar

- Slide
- Fade
- Blur

Top Navigation

- Underline grows
- Active glow
- Glass blur

Breadcrumb

- Fade
 - Slide Left
-

Card Animations

Cards should appear to float.

Default

Elevation 2

Hover

TranslateY(-6)

Scale(1.02)

Glow

Shadow increase

Press

Scale(.98)

Button Motion

Hover

- Glow
- Elevation
- Gradient Shift

Press

- Scale
- Ripple
- Shadow reduction

Success

- Green pulse
 - Check animation
-

AI Motion

ArenaMind AI introduces unique AI animations.

AI Thinking

Instead of spinner

Display

- Pulsing neural nodes
- Gradient wave
- Animated dots

- Streaming response
-

AI Typing

Characters stream naturally.

Avoid:

Entire paragraph appearing instantly.

Preferred

Streaming

with

cursor

AI Confidence

Confidence bar animates

0%

↓

98%

Never appears instantly.

AI Recommendation

Recommendation cards

Slide

↓

Glow

↓

Expand



Reveal reasoning

AI Reasoning Timeline

Thinking



Searching



Reasoning



Predicting



Generating



Completed

Each step animates progressively.

Dashboard Motion

Every dashboard updates live.

Examples

KPIs

Count upward.

Charts

Animate smoothly.

Maps

Move continuously.

Heatmaps

Pulse.

Notifications

Slide.

Alerts

Glow.

Charts

Charts never snap.

Line Charts

Draw from left.

Area Charts

Grow upward.

Bar Charts

Rise.

Pie Charts

Rotate.

Heatmaps

Pulse.

Radar Charts

Expand.

Digital Twin Motion

The stadium is alive.

Default

Slow rotation.

Hover

Highlight.

Selection

Zoom.

Incident

Red pulse.

Emergency

Camera focus.

AI Prediction

Animated overlay.

Crowd Motion

Crowds represented as particles.

Particles

Flow

↓

Congest

↓

Separate

↓

Redirect

Animated continuously.

Transport Motion

Vehicles

Move along routes.

ETA

Counts down.

Traffic

Flows continuously.

Parking

Updates smoothly.

Notification Motion

Critical

Shake

↓

Glow

↓

Expand

Warning

Slide

↓

Highlight

Info

Fade

↓

Appear

Modal Motion

Open

Scale

↓

Blur background

↓

Fade

Close

Reverse

Duration

250ms

Toast Motion

Enter

Top Right

↓

Slide

↓

Fade

Exit

Fade

↓

Slide

Search Motion

Focus

Expand

↓

Glow

↓

Suggestions

Typing

Suggestions update live.

Loading States

Never show

Loading...

Instead

Use

- Skeleton
 - AI Pulse
 - Digital Wave
 - Stadium Wireframe
 - Animated Heatmap
-

Empty States

Use illustrations.

Examples

No incidents.

The stadium is operating normally.



Hero Section Motion

Landing Page

Camera flies through stadium.

Fog moves.

Lights animate.

Crowd moves.

Digital Twin rotates.

AI nodes connect.

Particle network grows.

Everything should feel cinematic.

Scroll Animations

Scrolling should reveal

Section



Fade



Translate



Scale



Blur reduction

Avoid excessive effects.

Mouse Interactions

Cards

Tilt.

Buttons

Magnetic.

Hero

Parallax.

3D Stadium

Rotate.

Maps

Highlight.

Cursor

Custom cursor.

Glow.

Expands over

Buttons

Cards

Maps

3D Objects

Ambient Motion

Always active.

Examples

- Particle systems
- Floating nodes

- Neural connections
- AI pulse
- Stadium lights
- Weather
- Digital grid
- Animated gradients

Subtle.

Never distracting.

Accessibility

Users preferring reduced motion should receive:

- No parallax
- No camera movement
- Instant transitions
- Static particles
- Reduced animations

Respect

```
prefers-reduced-motion
```

Motion Performance

Target

60 FPS minimum

120 FPS preferred

GPU accelerated.

Avoid

- Layout thrashing
 - Excessive DOM updates
 - Heavy CSS filters
 - Blocking animations
-

Implementation Stack

Feature	Technology
Page Transitions	Framer Motion
Complex Timelines	GSAP
3D Motion	React Three Fiber
Camera Animation	Drei
Particle Systems	Three.js
Lottie Icons	Lottie React
Scroll Effects	GSAP ScrollTrigger
Physics	React Spring (optional)

Motion Checklist

Every screen should include:

-  Meaningful entrance animation
-  Hover feedback
-  Focus feedback
-  Loading animation
-  Success state
-  Error state
-  AI thinking state

-  Responsive transitions
 -  Accessibility support
 -  60 FPS performance
-

Motion Examples

AI Response

User Prompt

↓

AI Pulse

↓

Thinking Animation

↓

Streaming Response

↓

Confidence Meter

↓

Suggested Actions

Incident Detection

Camera detects anomaly

↓

Incident Card appears

↓

Map highlights location

↓

Notification slides in

↓

AI Recommendation expands

↓

Operator acknowledges

Crowd Prediction

Live Crowd Data

↓

Heatmap pulses

↓

AI predicts congestion

↓

Risk overlay appears

↓

Suggested alternate route animates

↓

Digital signage updates

Future Motion Enhancements

- Spatial UI transitions (Vision Pro style)
- Hand gesture interactions

- Voice-triggered animations
 - AI avatar micro-expressions
 - Holographic overlays
 - Live drone path animations
 - Stadium replay timeline
 - Multi-user collaborative cursors
-



Arena Motion System

"Motion should explain intelligence, not distract from it."

Every animation should make ArenaMind AI feel like the living operating system of the FIFA World Cup.